

MT1401

(Quiz + Midterm)

1) which of following is an implicit solution of the differential eq $\frac{dy}{dx} = -\frac{x}{y}$

(a) $x^2 + y^2 - 4 = 0$ (b) $x + y + 4 = 0$

$\frac{dy}{dx} = -\frac{x}{y}$ separable
 $\int y dy = \int -x dx$
 $y^2 = -x^2 + C \Rightarrow x^2 + y^2 = C$ (a)

2) The diff equation $\left(\frac{dx}{dy}\right) + \frac{1}{y}x = 2 \sin y$ is first order linear in unknown ...

(a) Variable x (b) Variable y

$\frac{dx}{dy}$ → dependent variable
 $\frac{dy}{dy}$ → independent variable
linear

3) which of following function would satisfy $\frac{dy}{dx} = \frac{dy}{dx^2} = \dots = \frac{dy}{dx^n}$

(a) $y = Ae^x$ (b) $y = Ane^x$

$\frac{dy}{dx} = Ae^x$
 $\frac{d^2y}{dx^2} = Ae^x$... same as $\frac{dy}{dx}$

4) Separable form of diff eq $\frac{dy}{dx} = y - 1$ is where

$v = y - 1$

(a) $\frac{dv}{dv} = dx$ (b) $\frac{dv}{v} = \frac{dx}{x}$

$v = y - 1$ w.r.t (a)
 $\frac{dv}{dx} = \frac{dy}{dx} \Rightarrow \frac{dy}{dx} = y - 1$
 $\frac{dy}{dx} = v + 1 - 1 = v$
 $d \frac{dv}{dx} = v = \frac{dv}{v} = dx$

5) General solution of separable diff eq $\frac{\sec y}{\tan y} dy = dx$ is

$y = \tan^{-1}(Ce^x)$

$\frac{\sec y}{\tan y} dy = dx \Rightarrow \int \frac{\sec y}{\tan y} dy = \int dx$
 $\int \frac{\sec y}{\tan y} dy = \int \frac{1}{\sin y \cos y} dy = \int \frac{1}{\sin y} dy = \ln|\tan y| = x + C$
 $\ln|\tan y| = x + C \Rightarrow \ln|\tan y| - C = x$
 $\ln\left(\frac{\tan y}{C}\right) = x \Rightarrow \frac{\tan y}{C} = e^x$
 $\tan y = Ce^x \Rightarrow y = \tan^{-1} Ce^x$

6) which of following substitution will transform the diff eq.

$\frac{dy}{dx} = \frac{x + y + 1}{x + y - 1}$ into separable form.

(a) $z = x + y$ (b) $z = y = vx$

$\frac{a_1}{a_2} = \frac{b_1}{b_2} = 1$ (Case 1)
 $z = x + y$ (value $a_2 b_1 = 1$)

(7) which following would be general solution of differential equation $\frac{dy}{dx} = 4$?

→ $y = 4x + c$

(8) $f(x, y) = \frac{2x}{3y} + 7$, $f(tx + ty) = \dots$

→ $f(x, y)$

(9) degree of the differential equation $\frac{dy}{dx} + 5 \left(\frac{dy}{dx} \right)^3 = e^{\sin x}$ is

→ 3

(10) if $x^3 dx + y^3 dy = 0$ has the equivalent form as $d\left(\frac{1}{3}x^3 y^3\right) = 0$ then its solutions

→ $x^3 y^3 = c$

(11) which of following is an initial value problem (IVP)

→ $\frac{d^2 y}{dx^2} + y = 0$, $y(1) = -2$ and $y'(1) = -2$

(12) differential equation $(x + 2y^3) \frac{dy}{dx} = y$ is linear unknown

→ Variable x ?

(13) differential equation $\frac{dx}{dy} + \frac{1}{y}x = 2 \sin y$ is first order linear unknown

→ Variable x

separable form $f(y)dy + g(x)dx = 0$ the differential equation $x \sin y dx + (x^2 + 1) \cot y dy = 0$ is

$$\Rightarrow \cot y dy + \frac{x}{x^2 + 1} dx = 0$$

if $d\left(\frac{x^2}{y}\right) = 0$ is an equivalent form of exact differential equation $2xy dx - x^2 dy = 0$ then the solution is

$$\Rightarrow x^2 = cy$$

which of following would be particular solution of differential equation $\frac{dy}{dx} = 4$

$$\Rightarrow y = 4x + a \quad \text{where } a \text{ is constant}$$

diff eq of form $\frac{dy}{dx} = f(x, y)$ is called separable if it can be written in the form

$$\Rightarrow \frac{dy}{dx} = h(x)g(y) \quad \left\{ \begin{array}{l} \text{Step 1: } \frac{dy}{dx} = h(x)g(y) \\ \text{Sep: } \frac{dy}{g(y)} = h(x)dx \\ \text{Int: } \int \frac{1}{g(y)} dy = \int h(x) dx \\ g(y) = H(x) + C \end{array} \right.$$

No need to use two constant of integration because

$$\Rightarrow C_1 - C_2 = C \quad C_1 + C_2 = C$$

following is an example of homogeneous function?

$$\Rightarrow f(x, y) = \sin\left(\frac{x}{y}\right) \quad \left\{ \begin{array}{l} f(x, y) = \sin\left(\frac{x}{y}\right) \\ f(tx, ty) = \sin\left(\frac{tx}{ty}\right) \end{array} \right.$$

differential equation $(\sin 2x - \tan y) dx - x \sec^2 y dy = 0$ is exact

$$\Rightarrow \frac{\partial M}{\partial x} = \frac{\partial N}{\partial y} = -\sec^2 y \quad \left\{ \begin{array}{l} M = \sin 2x - \tan y, \quad N = -x \sec^2 y \\ \frac{\partial M}{\partial y} = -\sec^2 y, \quad \frac{\partial N}{\partial x} = -\sec^2 y \\ \frac{\partial M}{\partial y} = \frac{\partial N}{\partial x} = -\sec^2 y \end{array} \right.$$

if the general solution of a separable differential equation $\sin^{-1}y = \cos^{-1}x + C$ provide that $y(\frac{1}{\sqrt{2}}) = \frac{1}{\sqrt{2}}$ then $C =$

$$\begin{cases} \sin^{-1}y = \cos^{-1}x + C \\ x = \frac{1}{\sqrt{2}}, y = \frac{1}{\sqrt{2}} \text{ Put in (1)} \\ \sin^{-1}(\frac{1}{\sqrt{2}}) = \cos^{-1}(\frac{1}{\sqrt{2}}) + C \\ \frac{\pi}{4} = \frac{\pi}{4} + C \\ 0 = C \end{cases}$$

The differential equation $\frac{dy}{dx} = \frac{y}{x}$ is

exact

$$\frac{dy}{dx} = \frac{y}{x} = \frac{1}{x} \cdot y$$

For non-exact diff equation $(x^2y - 2xy^2)dx - (x^3 - 2x^2y)dy = 0$

if $xM(x,y) + yN(x,y) = x^2y^2 \neq 0$ integration factor is

$$\Rightarrow \frac{1}{x^2y^2}$$

$$\begin{cases} \text{Case 1} \\ xM(x,y) + yN(x,y) \neq 0 \text{ then} \\ u = \frac{1}{xM(x,y) + yN(x,y)} \\ u = \frac{1}{x^2y^2} \end{cases}$$

On comparing the differential equation $dy = dx$

with standard form of $M(x,y)dx + N(x,y)dy$, we

$$\Rightarrow |M(x,y)| = |N(x,y)|$$

$\frac{dy}{dx} + 5\left(\frac{dy}{dx}\right)^3 - 3y = e^{\sin x}$ is an example of

differential equation.

Ordinary non-linear

(26) For $f(x, y) = x^2 - y^2$, $f(tx, ty) =$
 $\left. \begin{aligned} \text{put } (x, y) &\rightarrow (tx, ty) \\ f(tx, ty) &= t^2 x^2 - t^2 y^2 \\ &= t^2 (x^2 - y^2) = t^2 f(x, y) \end{aligned} \right\}$
 $\Rightarrow t^2 f(x, y)$

(27) For non-exact differential equation
 $y dx - (y - 3x - 3) dy = 0$, if $\frac{\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}}{M} = \frac{1}{y}$
then integration factor is
 $\Rightarrow y$
 $\left\{ \begin{aligned} &\text{if } \frac{\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}}{M} \text{ is function of } y \\ &U = \exp\left(\int \frac{\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}}{M} dy\right) \\ &U = e^{\int \frac{1}{y} dy} = e^{\ln y} = y \end{aligned} \right.$

(28) If $x^2 y dx + x^3 y^2 dy = 0$ has differential form
as $d(\frac{1}{3} x^3 y^3) = 0$ then its solutions are
 $\Rightarrow x^3 y^3 = C$
 $\left\{ \begin{aligned} &d(\frac{1}{3} x^3 y^3) = 0 \\ &\int d(\frac{1}{3} x^3 y^3) = \int 0 \\ &\frac{1}{3} x^3 y^3 = C_1 \\ &x^3 y^3 = 3C_1 \Rightarrow x^3 y^3 = C \end{aligned} \right.$

(29) General solution of non-homogeneous equation is
 $y_p + y_c$
 $\left\{ \begin{aligned} &\text{if } y_c \text{ is zero } \rightarrow \text{equation} \\ &y_p \text{ is particular + complementary solution} \\ &y_c \text{ is general solution} \end{aligned} \right.$

(30) The logistic equation $\frac{dy}{dt} = p(a - by)$ can be
identified as a non-linear equation that
 $\Rightarrow$ separable

Orthogonal trajectories occur naturally in the area of _____
Physics (b) electricity and magnetism (c) fluid dynamics

② differential equation $\frac{d^3y}{dx^3} + y^2 \neq x^4$ is _____
Nonhomogeneous & Nonlinear
 $\left\{ \begin{array}{l} \text{Homogeneous & zero p.d term} \\ \text{Linear} \end{array} \right\}$ $\left\{ \begin{array}{l} \text{one, 2, 3, 4} \end{array} \right\}$

③ Wronskian $W(h, z)(x) = 0$ then

$\Rightarrow h(x)$ and $z(x)$ are linear independent

④ homogeneous linear differential equation always possess the _____ solution.

$\Rightarrow$ Trivial

⑤ differential equation $\frac{d^2y}{dx^2} - 2xy + \sin x = 0$ is

$\Rightarrow$ Linear and homogeneous

⑥ Modification of non-linear logistic differential equation.

$\frac{1}{P} \cdot \frac{dP}{dt} = a - bP$ has been used in study

⑦ Solid tumor (b) revenue growth of commercial product

(c) actuarial prediction

⑧ Two function $f(x) = \pi$, $g(z) = \sin^2 x + \cos^2 x$ are

$\Rightarrow$ Linear dependent

⑨ Two families of curves are _____ at point of intersection.

$\Rightarrow$ Orthogonal

(39) leading coefficient in differential equation
 $3x \frac{dy}{dx} - 7x \frac{dy}{dx} + 5y = 0$
 $\Rightarrow 3$ higher derivative leading

(40) A function f is said to be analytic at a point a if it can be represented by a power series in $(x-a)$ with a positive radius of convergence.
 $\Rightarrow$ Analytical function

(41) logistic equation $\Rightarrow \frac{1}{p} \frac{dp}{dt} = a - bp$ in term
 $-bp^2$ can be interpreted as
 $\Rightarrow$ inhibition term { b is constant of proportionality term $-bp^2$, $b > 0$ can be inhibition term. }

(42) Wronskian of 1 and x^2 , $w(1, x^2) = \begin{vmatrix} \text{function} & 1 & x^2 \\ \text{derivative} & 0 & 2x \end{vmatrix} = 2x - 0 = 2x$
 $\Rightarrow 2x$

(43) let $f_1(x) = e^{mx}$ and $f_2(x) = e^{nx}$ are linear independent because $C_1 f_1(x) + C_2 f_2(x) = 0$
 $\Rightarrow C_1 = C_2 = \dots = C_n = 0$, m, n { $f_1(x), f_2(x), \dots, f_n(x)$ said to be linear independent on interval I if $C_1 f_1(x) + \dots + C_n f_n(x) = 0$ }

Q. The auxiliary equation of differential equation
 $y''' + 2y'' - 3y' = 0$ is _____

$$m^3 + 2m^2 - 3m = 0$$

$$\begin{cases} 2y'' - 5y' - 3y = 0 \\ (2m^2 - 5m - 3)e^{mx} = 0 \\ \begin{cases} y = e^{mx} \\ y' = me^{mx} \\ y'' = m^2 e^{mx} \end{cases} \end{cases}$$

auxiliary equation

Wronskian $W(1, e^x) =$
 $= e^x$

$$\begin{vmatrix} 1 & e^x \\ 0 & e^x \end{vmatrix} = e^x - 0 = e^x$$

Q. General solution of separable differential equation

$$\frac{dy}{1+y^2} = 2x dx$$

$$\begin{cases} \frac{1}{1+y^2} dy = 2x dx \\ \tan^{-1} y = x^2 + C \\ y = \tan(x^2 + C) \end{cases}$$

$$y = \tan(x^2 + C)$$

Q. If 2, 3, 5 are real roots of differential equation

then the general equation is _____

$$\begin{cases} m_1 = 2, m_2 = 3, m_3 = 5 \\ y = C_1 e^{2x} + C_2 e^{3x} + C_3 e^{5x} \\ y = C_1 e^{2x} + C_2 e^{3x} + C_3 e^{5x} \end{cases}$$

$$y = C_1 e^{2x} + C_2 e^{3x} + C_3 e^{5x}$$

Q. Order and degree of differential equation

$$1 + \frac{dy}{dx} + \frac{1}{2!} \left(\frac{dy}{dx} \right)^2 + \frac{1}{3!} \left(\frac{dy}{dx} \right)^3 + \dots$$

degree ∞

are one and undefined respectively

(4) Separable form $g(x)dx + f(y)dy = 0$
 of the differential eq $(2x \cos y)dx + x^2(\sec y - \sin y)dy = 0$

$$\Rightarrow \frac{2}{x}dx + (\sec y - \tan y)dy = 0$$

$$\begin{aligned} 2x \cos y &= -x^2(\sec y - \sin y)dy \\ \frac{dx}{x^2}dx &= -\frac{\sec y - \sin y}{\cos y}dy \\ \frac{2}{x}dx &= -(\sec y - \tan y)dy \\ \frac{2}{x}dx &= -\sec y - \tan y dy \end{aligned}$$

(5b) which of following is an equivalent form of exact differential equation.

$$y dx + x dy = 0$$

$$\begin{cases} x dy + y dx = 0 \\ d(xy) = 0 \end{cases}$$

$\Rightarrow d(xy) = 0$

(5i) On the general solution of a separable differential equation $\sin^{-1}y = \cos^{-1}x + C$, provided that $y(\frac{1}{\sqrt{2}}) = \frac{1}{\sqrt{2}}$ then

$\Rightarrow 0$

$$\begin{cases} \sin^{-1}y = \cos^{-1}x + C \\ \sin(\frac{1}{\sqrt{2}}) = \cos^{-1}(\frac{1}{\sqrt{2}}) + C \\ 45 = 45 + C \\ C = 0 \end{cases}$$

(52) A set of function $\{f_1(x), f_2(x), f_3(x), \dots, f_n(x)\}$ is said to be linearly independent on interval if there exist constant

$\Rightarrow$ Linear dependent

(53) differential equation $\frac{dy}{dx} = \frac{y}{x}$ is

$\Rightarrow$ Not exact

$$\begin{cases} \frac{1}{y}dy = \frac{1}{x}dx \\ \frac{1}{x}dx - \frac{1}{y}dy = 0 \\ M = \frac{1}{x} = \frac{\partial M}{\partial y} = 0 \\ N = -\frac{1}{y} = \frac{\partial N}{\partial x} = 0 \\ \Rightarrow \text{Not exact} \\ \Rightarrow \text{Linear} \end{cases}$$

Q1 If $d(\ln(xy)) = 0$ is an equivalent form of diff equation $xdy + ydx = 0$ then its solution is not

Q2 For a system in simple harmonic motion which of the following is the time required to complete a cycle of motion?

→ Period

Q3 A function satisfying the differential equation on interval I whose graph passes through (x_0, y_0) such that the slope of curve at point is the number y'_0 is called solution of the Value Problem.

→ initial

Q4 Wronskian of function $y_1(x) = 1, y_2(x) = \sin x, y_3(x) = \cos x$ is —

→ -1

Q5 Wronskian $w(1, e^x) = \text{—}$

→ e^x

Q6 The input function $g(x)$ can have one of the following forms. A constant function, A polynomial function, A trigonometric function, A rational function.

Q7 (a) exponential function e^{ax} (b) finite sum and products (c) Trigonometric function $\sin ax, \cos ax$

(60) The root of auxiliary equation $am^2 + bm + c = 0$ for differential equation $ay'' + by' + cy = 0$ is

$$m_1 = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

if $b^2 - 4ac > 0$ then $m_1 = \frac{-b + \sqrt{b^2 - 4ac}}{2a}$ and $m_2 = \frac{-b - \sqrt{b^2 - 4ac}}{2a}$

(61) differential equation

$$4x^2 \frac{dy}{dx} + 12 \frac{dy}{dx} + 3y = 0 \quad \text{subject of}$$

equation condition $y(4) = \frac{1}{8} \quad y'(4) = \frac{3}{64}$

⇒ Initial value problem

(62) a function satisfying the differential equation on the same interval I , containing a and b whose graph passes through two points

(a, y_0) and (b, y_1) is called solution value

⇒ boundary

(63) we can derive a differential equation governing the motion of mass attached to spring when the newton's second law combined with

⇒ Hooke's law

$$y = 2 + x \quad | \quad D^2(2+x) \\ \Rightarrow D^2$$

INITIAL
SHOH

- 64) combination of Newton's second law and Hooke's law could lead to differential equation governing to the motion of a mass attached to spring i.e. ——— motion.
 $\Rightarrow$ Simple harmonic motion.

- 65) roots of auxiliary equation $m^2 - 10m + 25(m-5)^2 = 0$ for differential equation is ———

$$\Rightarrow m_1 = m_2 = 5$$

- 66) annihilator operator for function $y(x) = 2e^{-x}$ ———
 $\Rightarrow D + 2$

- 67) annihilator operator for the function $y(x) = \frac{x^5}{5}$ ———
 $\Rightarrow D$

if $y = \frac{x^5}{5}$ derivative is $y' = x^4$
 $\therefore D = 1 \cdot 4 \cdot 3 \cdot 2 \cdot 1 = 24$

$D+1(5)$
$D(5)+5$
$0+5=5$

- 68) if $y = e^{5x}$ following is true.

- $\Rightarrow$ annihilator operator is $(D-5)$ $\leftarrow (D-5)'$

$(D-\alpha)^n$	$(D-5)(e^{5x})$
$\alpha = 5$	$e^{5x} \cdot 5 - 5 \cdot e^{5x} = 0$
$n = 1$	

- 69) Consider the non-homogeneous linear differential equation $a_n y^{(n)} + a_{n-1} y^{(n-1)} + \dots + a_1 y' + a_0 y = g(x)$

if the input function $g(x) = 5x + 7$,

Assume particular solution y_p could be ———

$$\Rightarrow Ax + B$$

$\Rightarrow (A_n + B) e^{5n}$
 Consider non-homogeneous linear equation
 $a_n y^{(n)} + a_{n-1} y^{(n-1)} + \dots + a_1 y' + a_0 y = g(x)$ if input function
 $g(x) = 5x^2 \sin 4x$, the assumed solution is

$$\Rightarrow (A_1 x^2 + B_1 x + C_1) \cos 4x + (A_2 x^2 + B_2 x + C_2) \sin 4x$$

(71) displacement measured below the equilibrium position are _____

$\Rightarrow$ positive

(72) Suppose that $y_1, y_2, \dots, y_n$ are solutions of the homogeneous n th differential equation.
 Then the solution is linearly dependent on interval if and only if

$$\Rightarrow W(y_1, y_2, \dots, y_n) = 0$$

(73) Auxiliary equation of differential equation

$$y''' + 2y'' - 3y = 0 \quad \text{is } \underline{\hspace{2cm}}$$

$$\Rightarrow m^3 + 2m^2 - 3 = 0$$

(74) Gravitational force exerted by the earth on body of mass m is called $\underline{\hspace{2cm}}$

$\Rightarrow$ weight

$$F_w = mg$$

$y = C_1 e^{(-2+\sqrt{6})x} + C_2 e^{(-2-\sqrt{6})x}$ is the complementary
 $y'' + 4y' - 2y = 2x^2 - 3x + 6$ General form of particular JP
 $\Rightarrow Ax^2 + Bx + C$

Q6 How frequency and period related in simple harmonic motion?

$\Rightarrow$ Inversely related

Q7 Vibration of an object on equilibrium point is called simple harmonic motion when the restoring force is proportion to _____

$\Rightarrow$ Spring Constant

Q8 annihilator operation function $g(x) = 8e^{3x}$

$\Rightarrow (D-3)$

$(D-\alpha)^n$

$(D-3)' = D-3$

Q9 For 2nd order linear non-homogeneous differential equation

$$a_2(x) \frac{d^2 y}{dx^2} + a_1(x) \frac{dy}{dx} + a_0(x)y = g(x)$$

which is not pair of boundary condition _____

$\Rightarrow y'(a) = y'_0, \quad y'(b) = y'_0$

Q10 second order linear ODE, if y_1 and y_2 are its solution and further $w(y_1, y_2) = \frac{2}{x} \frac{w_2 - w_1}{x}$

Then $U_2 =$ _____

$\Rightarrow -e^{2x}/2x$

(81) differential operator L is called an annihilator operator of function f , if $y = f(x)$ sufficiently diff (b) function $L(y) = 0$
 $\Rightarrow$ (c) L is linear diff operator with constant coeff

(82) second order linear ODE, if y_1 and y_2 are its solution and further $w(y_1, y_2) = 1$, $w_2 = -1/5$
 then $u'_2 =$ _____
 $\Rightarrow -1/20$

(83) set of function $\{f_1(x), f_2(x), f_3(x), \dots, f_n(x)\}$
 is _____ if at least one of them can expressed combination of remaining.

$\Rightarrow$ linear dependent

(84) Wronskian $w(x, 3x) =$ _____

$\Rightarrow 0$

(85) annihilator operator L for function $f(x) = e^{3x}$ is

$\Rightarrow D - 3$

$$(D - \alpha)^n \Rightarrow D - \alpha$$

(86) auxiliary equation of diff eq. $3y''' + y'' + 10y' - 4y = 0$
 $\Rightarrow 3m^3 + 5m^2 + 10m - 4 = 0$

87 equation of free undamped motion is

$$\Rightarrow \frac{d^2x}{dt^2} + \frac{k}{m}x = 0$$

88 differential operator D possesses the Property
 $\Rightarrow$ Linearity

89 For non homogeneous differential equation,
the form of particular solution $y_p + y_{p2} + \dots + y_{pn}$
Linear combination of all linear function,
Generated by repeated diff of input function $g(x)$
 $\Rightarrow$ independent

90 annihilator operator function $g(x) = 4 \sin x$

$$\Rightarrow (D^2 + 1)$$

$$\begin{matrix} \text{Idn} \\ e^{i\alpha x} \cos \beta x \\ e^{i\alpha x} \sin \beta x \end{matrix} \quad \begin{matrix} \alpha = 0 \\ \beta = 1 \end{matrix} \quad \begin{matrix} \text{sin} \\ \text{cos} \end{matrix}$$

$$\left((D^2 - 2\alpha D + (\alpha^2 + \beta^2))^n \right) \quad \begin{matrix} \alpha = 0 \\ \beta = 1 \end{matrix} \quad \begin{matrix} \text{sin} \\ \text{cos} \end{matrix}$$

$$(D^2 - 2(0)D + (0^2 + 1^2))^1 \quad \begin{matrix} \alpha = 0 \\ \beta = 1 \end{matrix} \quad \begin{matrix} \text{sin} \\ \text{cos} \end{matrix}$$

$$(D^2 + 1)$$

91 Solution of the auxiliary equation determine
the $\dots$ of diff equation.

$\Rightarrow$ solution

92 Polynomial Linear equation diff operator
with constant coefficients

$\Rightarrow$ Commute

93 Annihilator oper $g(x) \sin x$ is

$$\Rightarrow D^2$$

(94) differential operator D transform a function into another function.
 $\Rightarrow$ differentiable

(95) method of variation of parameter the derivative of function $u_2 = \frac{w_2}{w}$

(96) if 2, 3, 5 are real roots of diff operator general equation is $y_c = C_1 e^{2x} + C_2 e^{3x} + C_3 e^{5x}$

(97) what is annihilator operator of function $y(x) = 8e^{3x}$
 $\Rightarrow D-3$

$$L(y) = 0$$

$$y = C_1 e^{\frac{dx}{du}} + C_2 e^{\frac{dx}{du}} + \dots + C_n e^{\frac{dx}{du}}$$

$$(D-\alpha)^n$$

$8e^{3x}$
 $\alpha = 3$
 $n = 1$
 $(D-3)^1 = D-3$

(98) annihilator operator of function $x^4 + 5$ is D^5
 $\Rightarrow D^5$ (3 zero annihilates x^4)

(99) Period time is given by $\frac{2\pi}{\omega}$
 $\Rightarrow \frac{2\pi}{\omega}$ (period of vibration)

$$T = \frac{2\pi}{\omega}$$

100 hours low state that the force F is proportional to the Elongation

They are Inversely related

$$f = \frac{1}{T} = \frac{\omega}{2\pi}$$

$$\Rightarrow \frac{d^2 x}{dt^2} + \frac{k}{m} x = 0$$

(C) following is diff equation. $\frac{dy}{dx} = \frac{x^2 + y^2 + 1}{x^2 + 2y^2 + 1}$ separable.

$$x = \cancel{x} + h, \quad y = y + k$$

$$\frac{1}{j} + \frac{1}{2}$$

(14) $q_n(u) = 0$ in the differential equation

→ May not exist or may not be unique if $\lambda = 0$ / highest coefft.

Q5) following example of partial ^{diff} equation.

$$\frac{\partial n}{\partial y} = \frac{\partial y}{\partial n}$$

(166) following first order linear equation is
unknown variable y ?

← C.M

$$\Rightarrow x \frac{dy}{dx} + (\sin x) y = \cos x$$